

# Yihan Li

liyihan.xyz@gmail.com | +86 18615276438 | liyihan.xyz | github.com/HY-LiYihan

## Education

### Sun Yat-sen University

Sep 2023 – Jul 2027 (Expected)

*B.A. in English Language & Literature, School of Foreign Languages*

*Major GPA: 3.8/4.0*

*Minor in Intelligent Unmanned Systems*

*GPA: 4.0/4.0*

*Minor in Computer Science & Technology*

*GPA: 3.8/4.0*

*Selected CS coursework: Computer Technology 95; Mathematical Modeling 92; MATLAB Simulation 90; DSA Lab 89; NLP 86.*

## Publications

### Anticipating Contact for Visuo-Tactile Diffusion Policies

Under Review

Yating Feng, Yuhang Zheng, Lixin Xu, **Yihan Li**, Jiaju Yin, Renjing Xu

### Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation

Under Review

**Yihan Li**, Hanyi Zhang, Xiaoxi Jiang, Man Guo

### Unordered Landmark Visual Navigation

ECCV 2026

Hao Ren, Junzhe Zhu, **Yihan Li**, Zetong Bi, Le Zheng, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Lu Qi, Hui Cheng

### OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models

IROS 2026

**Yihan Li**, Hao Ren, Zhenglan Jiang, Junzhe Zhu, Hui Cheng

### RAPID Hand: A Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Generalist Robot Autonomy

NeurIPS 2025

Zhaoliang Wan, Zetong Bi, Zida Zhou, Hao Ren, Yiming Zeng, **Yihan Li**, Lu Qi, Xu Yang, Ming-Hsuan Yang, Hui Cheng

## Professional Experience

### HCL Lab (HKUST(GZ))

Jan 2026 – Present

*Research Assistant*

*Prof. Renjing Xu*

- Researching tactile sensing, vision models, VLA fine-tuning, VLA-based online RL, and Real2Sim scene reproduction for dexterous manipulation.
- Exploring multimodal visuo-tactile policy learning and evaluation protocols for embodied manipulation under real-world sensing constraints.

### Yun Drone

Feb 2025 – Present

*UAV Development Intern*

- Worked on UAV autonomy development across propulsion selection, formation flight, navigation and localization, and flight-test workflow design.
- Studied RL-based disturbance-resilient control and supported system iteration through flight-log analysis and control tuning.

### RAPID Lab (SYSU)

Nov 2024 – Mar 2026

*Research Assistant*

*Prof. Hui Cheng*

- Conducted embodied intelligence research across dexterous manipulation, multimodal perception, and vision-based navigation/traversability.
- Supported RAPID Hand, OmniTrav, and ULVN research directions through experimental validation, perception integration, and benchmark development.

## Leadership Experience

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### RoboTech Association (SYSU)

Sep 2024 – Dec 2025

*President; Competition Team Captain*

- Led robotics training and competition preparation across mechanical design, hardware, embedded control, vision, navigation, and algorithms.
- Coordinated association operations, technical mentoring, cross-group collaboration, and onboarding materials for 160+ members.

## Projects

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### Venom VNV: ROS 2 Unified Robotics Platform

Nov 2025 – Present

*Project Lead; ROS 2 Platform Developer*

[venom-algorithm.github.io/Venom\\_VNV](https://venom-algorithm.github.io/Venom_VNV)

- Architected a ROS 2 Humble platform spanning drivers, perception, localization, planning, mission control, simulation, and interface specifications for UGV/UAV/USV scenarios.
- Integrated Livox MID360, Hikvision cameras, AgileX chassis/arm/PX4 drivers with Point-LIO/Fast-LIO, Nav2 TEB, Ego Planner Swarm, YOLO detection, auto-aim tracking, and waypoint task management.

### RAPID Hand: Perception-Integrated Dexterous Manipulation Platform

Nov 2024 – Jun 2025

*Research Assistant*

[rapid-hand.github.io](https://rapid-hand.github.io)

- Co-developed a compact 20-DoF dexterous hand ontology and high-DoF teleoperation/data-collection stack for contact-rich whole-hand manipulation.
- Integrated hardware-level whole-hand perception, synchronizing wrist vision, fingertip tactile sensing, and proprioception with sub-7 ms latency for policy learning.

### OmniTrav: 360° Vision-based Traversability Benchmark

Jun 2025 – Mar 2026

*First Author*

- Proposed the C2T pipeline and built OmniTrav, a 4.7k-scene, 250k-frame, 577 GB dataset mapping pinhole, fisheye, and panoramic RGB to planner-ready 360-degree radial clearance.
- Designed an automated foundation-model annotation pipeline and a DINOv3-FPN baseline with a cross-attention angle projector for radial distance and obstacle prediction.

### GenreShift and Rosetta: LLM Systems for Linguistic Research

Oct 2024 – Present

*Project Leader*

[genreshift.cn](https://genreshift.cn)

[rosetta-stone.xyz](https://rosetta-stone.xyz)

- Built GenreShift for genre and discipline analysis, extracting quantitative linguistic features such as TTR, MTL, academic vocabulary, and cross-genre comparisons for LoRA/MoE-based genre transformation; filed invention patent application No. 2025105553019.
- Developed Rosetta as an agentic annotation pipeline with concept bootstrap, prompt optimization, embedding-based top-k retrieval, LLM judge routing, review feedback, and Prodigy-compatible JSONL export.

## Awards

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### China Robot and Artificial Intelligence Competition

2025

National First Prize (Team Captain)

### National Intelligent Unmanned Systems Application Challenge

2025

National Champion (Team Captain)

### National Undergraduate Electronics Design Contest

2025

National Second Prize (Team Captain)

### National Undergraduate Robotics Competition - RoboMaster University League

2025

National Second Prize (Vision Group Leader)

## Skills

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|---------------------------|----------------------------------------------------------------------|
| <b>Programming:</b>       | Python, C/C++, MATLAB, STM32/Arduino.                                |
| <b>Robotics and Sim:</b>  | ROS1/2, MuJoCo, Isaac Sim, Isaac Lab, Gazebo, PX4, Hermes, OpenClaw. |
| <b>Development Tools:</b> | Codex, Claude Code.                                                  |
| <b>Engineering Tools:</b> | Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot.     |
| <b>Languages:</b>         | Mandarin (native), English (TEM-4), French.                          |